World Happiness through the lens of the UN Sustainable Development and Financial Well-being

BUSINFO 702 Information Management, 2025 Quarter 3

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Executive Summary

This report provides a comprehensive analysis of the intricate relationship between World Happiness, Sustainable Development and Gross National Income. Three datasets were integrated: (i) Sustainable Development Report (SDG Index, 2000-2022), (ii) World Happiness Index (WHI, 2013-2023), and (iii) World Bank Gross National Income (2000-2023). This project designs and develops a data warehouse in SQLite where it analyses the relationship between these three datasets. Together, these provide a unique view of the influences of SDG and GNI on world happiness.

The star schema was built using a single fact table (country-year-indicator level) surrounded by four dimension tables for time, country, indicator and income. Three research questions guided the analysis: (i) Impact of SDG Index values on Happiness Index, (ii) Influence of GNI on Happiness Index, (iii) Nexus between SDG and GNI, denoting complex interplay and depicting divergence in happiness of nations.

Key findings reflect a strong positive correlation between the SDG Index and the happiness index. Income behaves as a dominant driver, and certain SDGs like Good Health and Well-being (SDG 3), Decent Work and Economic Growth (SDG 8), Industry, Innovation, and Infrastructure (SDG 9) provide a strong correlation with the happiness index.

The project exhibits how combining open datasets within a structured and well-defined data warehouse can provide global equity and sustainability insights. It also exemplifies the power of dimensional modelling and SQL for Business Analytics for real-world policy applications.

Research Questions

The following three research questions were used in this analysis:

Research Question 1 (R1)

- How strongly are a country's Happiness Index scores associated with its Sustainable Development Index (SDG) over time (2013-2022)?
- Significance: If the SDG Index and Happiness Index move in the same direction, this will strengthen the case and highlight whether progress on SDG increases the likelihood of happiness in the population.

Research Question 2 (R2)

- How does Gross National Income (GNI) impact the Happiness (WHI)?
- Significance: By validating GNI with happiness scores, we will be able to see if wealthier nations are consistently happier or if diminishing return exists.

Research Question 3 (R3)

- How do SGD progress and GNI level interact to explain the variation in the countries' Happiness Index Scores (2013-2023)? And which SDG goals matter?
- Significance: To analyse whether GNI, SDGs and World Happiness are not separate but parallel and are deeply connected for shaping equitable global development policies.

ELT Implementation in SQLite

A systematic Extract-Load-Transform (ELT) process was implemented in DB Browser to build the data warehouse. Three datasets were used: (i) Sustainable Development Report (SDG Index, 2000-2022), (ii) World Happiness Index (WHI, 2013-2023), and (iii) World Bank Gross National Income Classifications (2000-2023) – an additional dataset incorporated to enhance analysis with income groups. (Historical Classification by income link in the references section)

Extract (E)

The datasets were downloaded from Kaggle and the World Bank in CSV / Excel formats. A dedicated project folder structure was created to store the raw, staging, SQL and database files.

Load (L)

A new database stg_whi_dw.db was created. Three staging tables were created to load the raw data: (i) stg_sdr_raw for Sustainable Development Goals Data (ii) stg_whi_raw for World Happiness Data, and (iii) stg_gni_raw for World Bank Gross National Income Data. Data was imported directly into staging tables with minimal pre-processing to preserve integrity.

Transform (T)

The data was transformed as follows:

1. **Standardising Column names and Data Types**: The whitespaces in the text were trimmed, the years were cast into integers, and income categories were normalised to uppercase.
2. **Empty Strings to NULLs**: Empty spaces were converted to NULL for country, year, index values, SDG goal values, and income levels across all three datasets.
3. **Creating Long Tables**:
    a. SDG dataset – Each goal was originally a column; therefore, data was unpivoted into a new table stg_sdr_long for each row to represent country, year, indicator, value.
    b. WHI dataset – Split into two indicators (happiness_score and happiness_rank) and input into the stg_whi_long table.
4. **Attribute Hierarchies**: Two attribute hierarchies were created: (i) Time Hierarchy, where years were converted to form decades, and (ii) Country and Income Group (socio-economic hierarchy), where country is the most granular level (Eg: Indonesia, Finland etc.) and Income group is a higher-level grouping of countries (Low, Lower-Middle, Upper-Middle, High as defined by World Bank GNI)
5. **Indexing**: Indexes were created per table to speed the joining process.

ELT Validation and Final Output

After finishing with the ELT process, a final summary count was performed based on the intersection of the three datasets. The results are as follows:

1. 156 countries were consistently present across all three datasets
2. The overlapping time coverage was aligned from the year 2013 to 2022. For these years, the complete data was available across all three datasets.
3. A total of 1404 country-year records were present across all three datasets, forming the base for the SQL for Business Analytics queries.

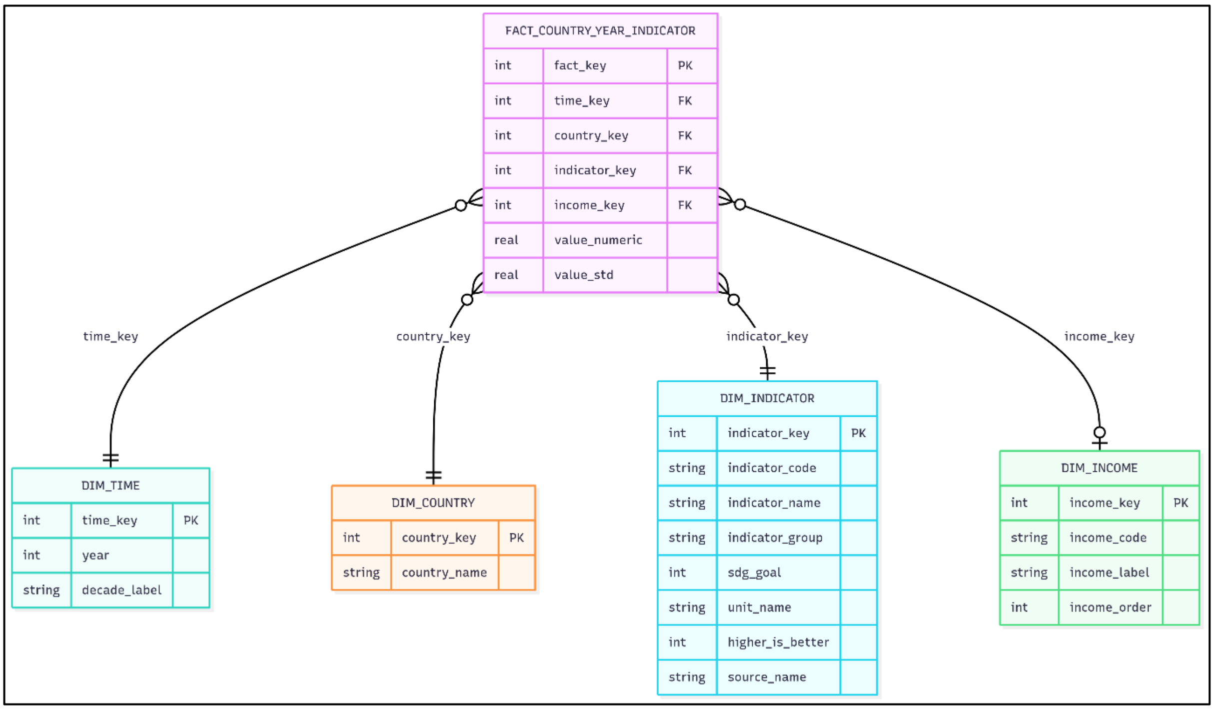
Star Schema Design

A 100% star schema design was created with one fact table (FactCountryYearIndicator) in the middle surrounded by four dimension tables (DimTime, DimCountry, DimIndicator and DimIncome). Refer to Figure 1. DimTime served as the time dimension for this star schema. The following are the details:

1. **Dimension Table Population**: Distinct values were inserted into the dimension tables for creation of attribute hierarchy.
    a. DimTime was created to hold year data, which was further mapped to create decades.
    b. DimCountry merged country names from all datasets.
    c. DimIncome was created to hold the income_code categories and their names.
    d. DimIndicator held the codes of SDG Index, SDG Goals, Happiness Index and Happiness Rank. It served as single dimension for all the measures.
2. **Fact Table Creation**: The central FactCountryYearIndicator table was populated using the four dimension tables. The primary keys of all dimension tables serve as foreign keys in the fact table. This helps in conforming dimensions and referential integrity. One record per country, year, indicator and income was created.

Clear naming conventions were established, dim_* prefix for dimension tables, fact_* prefix for fact table and *_key suffix for keys. The star schema unified all three datasets (SDG, WHI and GNI), which helped derive the correlations between SDG and Happiness Indexes (Research Question 1), Income groups and Happiness Index (Research Question 2) and combining both drivers in regression (Research Question 3).

Figure 1: ERD Diagram of Star Schema



SQL for Business Analytics

To address the three research questions, a series of SQL queries was implemented against the view built using the star schema.

Research Question 1: Relationship between SDG Progress and World Happiness

- **Query Logic**: Joined SDG Index values with Happiness Index values at the country-year level (For analysis purposes, only the latest common year, i.e. 2022, was considered). The Pearson correlation coefficient between the SDG Index and the Happiness Index was calculated for each common year (2013 – 2022). An overall correlation coefficient was also calculated using statistical formulas.

- **Findings**: The overall Pearson correlation coefficient was 0.748 (~0.75), suggesting a strong positive correlation between the two indices. Table 1 shows the year-by-year analysis of the correlation coefficient (between 0.69 and 0.81), indicating a strong relation and robustness across time between the two indices. For e.g.: Table 2 shows that Nordic Countries like Finland, Denmark, Sweden display high SDG scores (> 85) and are also high on the happiness index (> 7.3). On the other hand, Table 3 shows that countries like Afghanistan, Zimbabwe and Malawi score poorly on both indices due to governance challenges, economic hardships or conflicts.

- **Takeaway**: Aligning policies to improve SDGs is strongly associated with improved happiness.

Figure 2: SQL Statement for Pearson Correlation Calculation

```sql
--Pearson Correlation (Overall, all years)
WITH pairs AS (
  SELECT SDG_Index AS x, Happiness_Index AS y
  FROM vw_country_year_summary
  WHERE SDG_Index IS NOT NULL AND Happiness_Index IS NOT NULL
),
stats AS (
  SELECT COUNT(*) AS n,
         SUM(x) AS sx, SUM(y) AS sy,
         SUM(x*x) AS sxx, SUM(y*y) AS syy,
         SUM(x*y) AS sxy
  FROM pairs
)
SELECT (n*sxy - sx*sy) /
       NULLIF( sqrt((n*sxx - sx*sx)*(n*syy - sy*sy)), 0 ) AS pearson_r
FROM stats;

--correlation by year
WITH pairs AS (
  SELECT Year, SDG_Index AS x, Happiness_Index AS y
  FROM vw_country_year_summary
  WHERE SDG_Index IS NOT NULL AND Happiness_Index IS NOT NULL
),
stats AS (
  SELECT Year,
         COUNT(*) n, SUM(x) sx, SUM(y) sy,
         SUM(x*x) sxx, SUM(y*y) syy, SUM(x*y) sxy
  FROM pairs
  GROUP BY Year
)
SELECT Year,
       (n*sxy - sx*sy) /
       NULLIF( sqrt((n*sxx - sx*sx)*(n*syy - sy*sy)), 0 ) AS pearson_r
FROM stats
ORDER BY Year;
```

Figure 3: SQL Statement for Top 10 and Bottom 10 Happy Countries Calculation

```
--Top 10 happy countries
WITH latest AS (
  SELECT MAX(Year) AS y
  FROM vw_country_year_summary
  WHERE SDG_Index IS NOT NULL AND Happiness_Index IS NOT NULL
)
SELECT Country, Year, Happiness_Index, SDG_Index
FROM vw_country_year_summary, latest
WHERE Year = latest.y
  AND Happiness_Index IS NOT NULL
  AND SDG_Index IS NOT NULL
ORDER BY Happiness_Index DESC
LIMIT 10;

--Bottom 10 unhappy countries
WITH latest AS (
  SELECT MAX(Year) AS y
  FROM vw_country_year_summary
  WHERE SDG_Index IS NOT NULL AND Happiness_Index IS NOT NULL
)
SELECT Country, Year, Happiness_Index, SDG_Index
FROM vw_country_year_summary, latest
WHERE Year = latest.y
  AND Happiness_Index IS NOT NULL
  AND SDG_Index IS NOT NULL
ORDER BY Happiness_Index ASC
LIMIT 10;
```

Table 1: Output of Correlation between SDG Index and Happiness Index by year

	Year	pearson_r
1	2013	0.689552949297105
2	2015	0.724414803749892
3	2016	0.75159209003888
4	2017	0.77035353044241
5	2018	0.767558451811408
6	2019	0.766565105489353
7	2020	0.76104034799545
8	2021	0.764057149382093
9	2022	0.76343179236038

Table 2: Output of Top 10 Happy Countries and their SDG Index

	Country	Year	Happiness_Index	SDG_Index
1	Finland	2022	7.821	86.8
2	Denmark	2022	7.636	85.7
3	Iceland	2022	7.557	78.3
4	Switzerland	2022	7.512	80.5
5	Netherlands	2022	7.415	79.4
6	Luxembourg	2022	7.404	77.6
7	Sweden	2022	7.384	86.0
8	Norway	2022	7.365	82.0
9	Israel	2022	7.364	74.0
10	New Zealand	2022	7.2	78.4

Table 3: Output of Bottom 10 Happy Countries and their SDG Index

	Country	Year	Happiness_Index	SDG_Index
1	Afghanistan	2022	2.404	49.0
2	Lebanon	2022	2.955	67.5
3	Zimbabwe	2022	2.995	55.6
4	Rwanda	2022	3.268	60.2
5	Botswana	2022	3.471	62.7
6	Lesotho	2022	3.512	54.9
7	Sierra Leone	2022	3.574	55.7
8	Tanzania	2022	3.702	56.8
9	Malawi	2022	3.75	56.3
10	Zambia	2022	3.76	54.3

Research Question 2: Impact of Income and World Happiness

- **Query Logic**: Joined Gross National Income groups with Happiness Index values at the country-year level (For analysis purposes, only the latest common year, i.e. 2023, was considered). Group-wise happiness index averages were calculated and compared.
- **Findings**: Countries with higher income categories (High, Upper-middle) are much more likely to report higher happiness than low-income countries. Table 4 indicates that low-income countries (avg. happiness = 4.04) often struggle with basic needs and political/economic instability, thus lagging behind in comparison to high-income countries (avg. happiness = 6.5), which benefit from better governance, robust infrastructure, and social welfare. The high-income countries like Finland, Denmark, and Iceland have happiness indexes above 7, and low-income countries like Afghanistan, Sierra Leone, and Congo DR have happiness indexes <= 3. (Refer Table 5 and Table 6)
- **Takeaway**: The gap between the high-income and low-income countries is stark. This confirms that income is a strong determinant of happiness index, highlighting the importance of economic security for well-being.

Figure 4: SQL statement for Income Level wise Average Calculation

```sql
-- Latest shared year example: change/remove the Year filter as you like
SELECT Income_Level,
       AVG(Happiness_Index) AS avg_happiness,
       COUNT(*)             AS n_country_years
FROM vw_country_year_summary
WHERE Happiness_Index IS NOT NULL
GROUP BY Income_Level
ORDER BY CASE Income_Level
  WHEN 'Low income' THEN 1
  WHEN 'Lower middle income' THEN 2
  WHEN 'Upper middle income' THEN 3
  WHEN 'High income' THEN 4
END;
```

Figure 5: SQL Statement for Top 10 High-income countries by Happiness Index

```
--Top 10 High-Income Countries by Happiness Index (Latest Year)
SELECT
    c.country_name AS Country,
    t.year AS Year,
    f.value_numeric AS Happiness_Index,
    d.income_label AS Income_Level
FROM FactCountryYearIndicator f
JOIN DimCountry c   ON f.country_key = c.country_key
JOIN DimTime t      ON f.time_key = t.time_key
JOIN DimIndicator i ON f.indicator_key = i.indicator_key
JOIN DimIncome d    ON f.income_key = d.income_key
WHERE i.indicator_code = 'HAPPINESS_SCORE'
  AND d.income_label = 'High income'
  AND t.year = (SELECT MAX(year) FROM DimTime)
ORDER BY f.value_numeric DESC
LIMIT 10;
```

Figure 6: SQL Statement for Bottom 10 Low-income countries by Happiness Index

```
--Bottom 10 Low-Income Countries by Happiness Index (Latest Year)
SELECT
    c.country_name AS Country,
    t.year AS Year,
    f.value_numeric AS Happiness_Index,
    d.income_label AS Income_Level
FROM FactCountryYearIndicator f
JOIN DimCountry c   ON f.country_key = c.country_key
JOIN DimTime t      ON f.time_key = t.time_key
JOIN DimIndicator i ON f.indicator_key = i.indicator_key
JOIN DimIncome d    ON f.income_key = d.income_key
WHERE i.indicator_code = 'HAPPINESS_SCORE'
  AND d.income_label = 'Low income'
  AND t.year = (SELECT MAX(year) FROM DimTime)
ORDER BY f.value_numeric ASC
LIMIT 10;
```

Table 4: Output of Income Group-wise Average Happiness Index Scores

	Income_Level	avg_happiness	n_country_years
1	NULL	5.26798360655738	61
2	Low income	4.04274025974026	231
3	Lower middle income	4.85055405405405	370
4	Upper middle income	5.49481637717122	403
5	High income	6.58975109170306	458

Table 5: Output of Top 10 High-Income Countries by Happiness Index

	Country	Year	Happiness_Index	Income_Level
1	Finland	2023	7.804	High income
2	Denmark	2023	7.586	High income
3	Iceland	2023	7.53	High income
4	Israel	2023	7.473	High income
5	Netherlands	2023	7.403	High income
6	Sweden	2023	7.395	High income
7	Norway	2023	7.315	High income
8	Switzerland	2023	7.24	High income
9	Luxembourg	2023	7.228	High income
10	New Zealand	2023	7.123	High income

Table 6: Output of Bottom 10 low-income countries by Happiness Index

	Country	Year	Happiness_Index	Income_Level
1	Afghanistan	2023	1.859	Low income
2	Sierra Leone	2023	3.138	Low income
3	Congo, Dem. Rep.	2023	3.207	Low income
4	Malawi	2023	3.495	Low income
5	Madagascar	2023	4.019	Low income
6	Liberia	2023	4.042	Low income
7	Ethiopia	2023	4.091	Low income
8	Togo	2023	4.137	Low income
9	Mali	2023	4.198	Low income
10	Gambia, The	2023	4.279	Low income

Research Question 3: Impact of SDG Progress and Income (combined) on World Happiness

- **Query Logic**: A multiple regression model was implemented to check the effect of SDG Index and Income Groups on World Happiness. Income levels were assigned orders (1 being the lowest income group and 4 being the highest). The top 5 most correlated SDG goals (individual goal score) were computed using the statistical Pearson correlation coefficient formula. Refer Figure

- **Findings**: Table 7 suggests that the regression model gives a beta value of ~0.03 for the SDG Index, income had a beta value of ~0.57, and the intercept was 1.63. Income has a strong coefficient and thus explains a larger variation in happiness than the SDG index. Table 8 shows us the top 5 SDG goals most correlated with the happiness index. This indicates that certain SDGs (health, jobs, infrastructure, etc.) matter more for improving happiness than others (climate, energy, inequality, etc).

- **Takeaway**: This suggests that income is a dominant driver. From a policy perspective, for low-income countries, the focus should be on lifting incomes. SDGs are powerful for increasing happiness in the middle-income nations. For the high-income nations, SDGs maintain quality of life, but factors such as mental health, trust, etc., become a priority.

Figure 7: SQL Statement for Multiple Linear Regression

```sql
WITH base AS (
  SELECT
    SDG_Index AS x,                       -- predictor 1
    CASE Income_Level                     -- predictor 2 (numeric)
      WHEN 'Low income' THEN 1
      WHEN 'Lower middle income' THEN 2
      WHEN 'Upper middle income' THEN 3
      WHEN 'High income' THEN 4
    END AS z,
    Happiness_Index AS y
  FROM vw_country_year_summary
  WHERE SDG_Index IS NOT NULL AND Happiness_Index IS NOT NULL
        AND Income_Level IS NOT NULL
),
means AS (
  SELECT AVG(x) AS xbar, AVG(z) AS zbar, AVG(y) AS ybar FROM base
),
cov AS (
  SELECT
    SUM( (b.x - m.xbar)*(b.x - m.xbar) )              AS Sxx,
    SUM( (b.z - m.zbar)*(b.z - m.zbar) )              AS Szz,
    SUM( (b.x - m.xbar)*(b.z - m.zbar) )              AS Sxz,
    SUM( (b.x - m.xbar)*(b.y - m.ybar) )              AS Sxy,
    SUM( (b.z - m.zbar)*(b.y - m.ybar) )              AS Szy,
    COUNT(*)                                          AS n,
    m.xbar, m.zbar, m.ybar
  FROM base b, means m
)
SELECT
  -- Regression coefficients (closed-form for 2 predictors)
  (Sxy*Szz - Szy*Sxz) / (Sxx*Szz - Sxz*Sxz) AS beta_sdg,
  (Szy*Sxx - Sxy*Sxz) / (Sxx*Szz - Sxz*Sxz) AS beta_income,
  -- Intercept
  m.ybar - ((Sxy*Szz - Szy*Sxz) / (Sxx*Szz - Sxz*Sxz))*m.xbar
         - ((Szy*Sxx - Sxy*Sxz) / (Sxx*Szz - Sxz*Sxz))*m.zbar AS intercept,
  n AS rows_used
FROM cov, means m;
```

Figure 8: SQL Statement for Top 5 most correlated SDG Goals

```sql
WITH latest_common_year AS (
  SELECT MIN(max_y) AS y FROM (
    SELECT MAX(t.year) AS max_y
    FROM FactCountryYearIndicator f
    JOIN DimIndicator i ON i.indicator_key = f.indicator_key
    JOIN DimTime t      ON t.time_key      = f.time_key
    WHERE i.indicator_code = 'HAPPINESS_SCORE'
    UNION ALL
    SELECT MAX(t.year) AS max_y
    FROM FactCountryYearIndicator f
    JOIN DimIndicator i ON i.indicator_key = f.indicator_key
    JOIN DimTime t      ON t.time_key      = f.time_key
    WHERE i.indicator_group = 'SDG_GOAL'   -- SDG01..SDG17
  )
),
happiness AS (
  SELECT f.country_key, f.value_numeric AS y
  FROM FactCountryYearIndicator f
  JOIN DimIndicator i ON i.indicator_key = f.indicator_key
  JOIN DimTime t      ON t.time_key      = f.time_key
  WHERE i.indicator_code = 'HAPPINESS_SCORE'
    AND t.year = (SELECT y FROM latest_common_year)
),
sdg AS (
  SELECT f.country_key, i.indicator_code, i.indicator_name, i.sdg_goal,
         f.value_numeric AS x
  FROM FactCountryYearIndicator f
  JOIN DimIndicator i ON i.indicator_key = f.indicator_key
  JOIN DimTime t      ON t.time_key      = f.time_key
  WHERE i.indicator_group = 'SDG_GOAL'
    AND t.year = (SELECT y FROM latest_common_year)
),
pairs AS (
  SELECT s.indicator_code, s.indicator_name, s.sdg_goal, s.x, h.y
  FROM sdg s
  JOIN happiness h ON h.country_key = s.country_key
  WHERE s.x IS NOT NULL AND h.y IS NOT NULL
),
```

```sql
stats AS (
  SELECT indicator_code, indicator_name, sdg_goal,
         COUNT(*) n,
         SUM(x) sx, SUM(y) sy,
         SUM(x*x) sxx, SUM(y*y) syy, SUM(x*y) sxy
  FROM pairs
  GROUP BY indicator_code, indicator_name, sdg_goal
)
SELECT
  (SELECT y FROM latest_common_year) AS year,
  indicator_code,
  indicator_name,
  sdg_goal,
  (n*sxy - sx*sy) /
  NULLIF( sqrt((n*sxx - sx*sx) * (n*syy - sy*sy)), 0 ) AS pearson_r,
  n AS pairs_used
FROM stats
ORDER BY pearson_r DESC
LIMIT 5;
```

Table 7: Output of Regression Coefficient Table

	beta_sdg	beta_income	intercept	rows_used
1	0.0334989524408719	0.567563286626298	1.63605648869415	1311

Table 8: Output of Top 5 most correlated SDG Goals with Happiness Index

	year	indicator_code	indicator_name	sdg_goal	pearson_r	pairs_used
1	2022	SDG03	SDG 3: Good Health and Well-being	3	0.80143814894	138
2	2022	SDG08	SDG 8: Decent Work and Economic …	8	0.746743697494763	138
3	2022	SDG09	SDG 9: Industry, Innovation, and …	9	0.736695311898418	138
4	2022	SDG11	SDG 11: Sustainable Cities and …	11	0.735691564897307	138
5	2022	SDG06	SDG 6: Clean Water and Sanitation	6	0.729292747569562	138

Insight and Future Work

Based on the analysis from 2013-2023, it clearly depicts that a nation's well-being is not defined by a single metric but by a complex metric of economic, social and environmental factors. While GNI remains a foundational driver, but is not the only destination for happiness, particularly in wealthier nations where its influence diminishes. Happiness is propelled by non-economic factors similar to the SDGs, for example, the level of resilience displayed during COVID-19 despite low macro-economic indicators. In essence, a sustainable path to progress requires moving beyond a singular metric and using an integrated framework that equally prioritises between economic prosperity and social well-being. The future work is expected to focus on (i) financing for sustainable development, (ii) strengthening social capital, and (iii) policy development and implementation.

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